

Project Requirement Document

AI Lead Qualification System — Capstone Project (Option 3)

Petalnexus Pvt. Ltd. — AI Automation Internship
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Problem Statement

Sales teams manually review every inbound lead (from web forms, landing pages, etc.) to judge whether it is worth pursuing. This is slow, inconsistent, and delays response time to genuinely hot leads — sometimes losing them to competitors who respond faster.

Current Process & Pain Points

- Leads land in an inbox or spreadsheet with no automatic validation — duplicate, junk, or incomplete entries mixed in with real ones
- A person manually reads each lead and decides if it is worth a sales call — subjective, slow, and does not scale
- No consistent scoring criteria — one rep may pursue a lead another would ignore
- Hot leads can sit unnoticed for hours before anyone follows up
- No structured nurture path for leads that are not sales-ready yet — they are simply forgotten

Proposed Solution Outline

Trigger	Webhook receives lead submissions (name, email, company, message, etc.)
Input	Raw lead form data
Workflow	Validate → enrich (optional company/domain lookup) → AI qualification & scoring → store in DB → branch on score
AI Role	LLM evaluates lead quality/intent and assigns a qualification score with reasoning
Business Rules	Score thresholds define hot vs. warm vs. cold; only valid, non-duplicate leads proceed past validation
Data Storage	Supabase (Postgres) leads table — stores lead details, score, qualification status, and timestamps
Output / Notifications	Hot leads trigger an immediate sales alert email (Gmail SMTP); warm/cold leads enter a follow-up/nurture email flow

Success Criteria

- 100% of incoming leads are validated and stored with no manual data entry
- AI scoring reliably differentiates hot vs. cold leads on defined criteria
- Sales team receives hot-lead alerts within seconds of submission
- No lead is lost — every lead has a recorded status and next action (alert or nurture)